

Are galaxies in compact groups special?

Matthieu Tricottet[1⋆], Gary A. Mamon[2⋆⋆], and Eugenia Díaz-Giménez[3,4⋆⋆⋆]

[1] 80, rue d'Alésia, 75014 Paris, France
e-mail: `matthieu.tricottet@gmail.com`
[2] Institut d'Astrophysique de Paris (UMR 7095: CNRS & Sorbonne Université), 98 bis boulevard Arago, 75014 Paris, France
[3] CONICET. Instituto de Astronomía Teórica y Experimental (IATE), Laprida 854, X5000BGR, Córdoba, Argentina
[4] Universidad Nacional de Córdoba (UNC). Observatorio Astronómico de Córdoba (OAC), Laprida 854, X5000BGR, Córdoba, Argentina

ABSTRACT

Context. Compact groups combine high projected densities with low velocity dispersions, but many are embedded in larger structures. Their significance depends on whether their galaxies remain distinct from comparable regular-group members, not only from field galaxies.

Aims. We test whether galaxies in compact groups of four members (CG_4) remain special against regular groups of four members (RG_4), the four brightest galaxies of richer regular groups ($\mathrm{Control}_{4\mathrm{B}}$), and projected compact cores formed by the brightest group galaxy and its three nearest projected companions ($\mathrm{Control}_{4\mathrm{C}}$).

Methods. From our previous samples, we retain 62 CG_4 systems after removing split systems. We combine Galaxy Zoo 1 vote fractions, Sloan Digital Sky Survey stellar masses and specific star-formation rates, optical colours, emission-line diagnostics, and seeing-corrected half-light radii. Inference proceeds through four complementary analyses: per-control covariate-adjusted galaxy-level models, galaxy-level matching, group-level matching of satellite composition, and within-host stratified comparisons.

Results. The main residual signal is morphological and estimand-dependent. CG_4 galaxies, especially satellites, have higher elliptical-vote and lower spiral-vote fractions than their regular-group analogues. The adjusted individual-galaxy excess is strong against $\mathrm{Control}_{4\mathrm{B}}$ and RG_4 but not detected against $\mathrm{Control}_{4\mathrm{C}}$; in contrast, the matched satellite-composition contrast is positive against all three controls under the stated per-control convention, including $\mathrm{Control}_{4\mathrm{C}}$. No additional within-host excess is detected with the primary conditional estimator. The morphology result is stable to the vote-fraction threshold, a threshold-free vote-fraction model, a Sérsic-index cross-check, and the exclusion of the closest projected pairs. Conditioning on a projected pairwise tidal-density proxy attenuates the galaxy-level coefficient. Star-formation, colour, active-galactic-nucleus, magnitude-gap, Hα, and fixed-mass size diagnostics do not show comparably robust adjusted or matched CG_4 offsets.

Conclusions. CG_4 systems host a population shifted towards early-type-like morphologies, consistent with accumulated transformation in dense group cores and/or preprocessing rather than proof of a uniquely isolated present-day quenching mode.

Key words. galaxies: groups: general – galaxies: evolution – galaxies: interactions – galaxies: star formation – galaxies: elliptical and lenticular, cD – methods: statistical

1. Introduction

Given their high apparent densities and low velocity dispersions (Hickson et al. 1992), compact groups appear to be ideal sites for the frequent tidal encounters and mergers that alter galaxy masses, luminosities, sizes, morphologies, and star-formation rates (e.g., Mamon 1992). Mendes de Oliveira & Hickson (1994) provided the classic report of an excess of early-type galaxies in compact groups relative to the field, in line with the broader morphology–density relation (Dressler 1980), and multi-wavelength studies of compact-group galaxies likewise report altered infrared, UV, star-formation, and stellar-population properties relative to looser environments (Johnson et al. 2007; Tzanavaris et al. 2010; Bitsakis et al. 2011; Coenda et al. 2012).

The interpretation of such contrasts is nevertheless ambiguous. Projected selection can include chance alignments, and many Hickson-like systems need not be isolated bound entities: apparent compact groups may be transient alignments, dense substructures, or cores embedded in looser groups and richer haloes (Mamon 1986; Díaz-Giménez & Mamon 2010; Díaz-Giménez et al. 2020; Zheng & Shen 2021; Taverna et al. 2023). The relevant observational question is therefore not only whether compact-group galaxies differ from field galaxies, but whether they remain different from galaxies occupying regular-group analogues selected with comparable luminosity, redshift, and membership constraints. Recent structural studies point to transformation in progress in compact groups, with stronger signatures in non-isolated systems and host struc-

⋆ ORCID: https://orcid.org/0009-0008-0062-4450
⋆⋆ ORCID: https://orcid.org/0000-0001-8956-5953
⋆⋆⋆ ORCID: https://orcid.org/0000-0001-5374-4810

tures (Montaguth et al. 2023, 2025, 2026a,b), but they do not by themselves answer this control question.

In our previous work (Tricottet et al. 2025), we found that, once the direct consequences of the compact-group selection are accounted for, compact groups of four galaxies have group-scale properties broadly similar to those of regular groups of four galaxies and to the cores of richer regular groups. Here we ask the complementary question: even if compact groups are not strongly distinct as systems, are their member galaxies individually different? We compare CG_4 member galaxies with three regular-group analogues – RG_4, $\mathrm{Control}_{4\mathrm{B}}$, and $\mathrm{Control}_{4\mathrm{C}}$ (Sect. 2.1) – controlling for stellar mass, redshift, galaxy rank, and available group-scale properties, and use complementary individual-galaxy, group-composition, and shared-host contrasts to identify which observable, if any, carries a residual compact-group signal.

Section 2 describes the samples, the classifications, and the statistical design. Section 3 presents the population contrasts, the matched and within-host comparisons, and the robustness and secondary diagnostics. Section 4 discusses the physical interpretation, and Section 5 concludes.

2. Data and classifications

2.1. Samples

We use the compact-group and control samples constructed in our previous study (Tricottet et al. 2025), whose construction we briefly recall. Galaxy properties are taken from the SDSS DR12 catalogue of Tempel et al. (2017), while regular groups are drawn from the SDSS DR7 group catalogue of Lim et al. (2017); the two catalogues were matched through SDSS object identifiers to define the Lim–Tempel working catalogue. Compact groups are selected from the modified Hickson compact-group catalogue of Díaz-Giménez et al. (2018): reliable HMCGs with exactly four members, all present in the Lim–Tempel catalogue, satisfying the volume- and luminosity-complete limits $0.005 \leq z_{\mathrm{BGG}} \leq 0.0452$ and $M_{r,\mathrm{BGG}} \leq -21.81$, with all four galaxies lying within 3 magnitudes of the brightest group galaxy (BGG). In contrast to our previous study, we remove CGs classified as *split* following Zheng & Shen (2021), which reduces the compact-group sample from 78 to 62 groups.

The regular-group comparison samples are built from Lim–Tempel groups satisfying the same redshift and BGG-luminosity limits, with at least 4 members and at least 3 satellites within 3 magnitudes of the BGG. The three controls serve different physical purposes. RG_4 contains regular groups with exactly four members (excluding systems identical to a CG_4) and is the most direct equal-membership comparison. $\mathrm{Control}_{4\mathrm{B}}$ contains the four brightest galaxies of each richer regular group and tests whether the CG_4 population resembles the luminous core of an ordinary group. $\mathrm{Control}_{4\mathrm{C}}$ contains the BGG plus its three closest projected companions and is the most compact-core-like ordinary-group analogue; it is regenerated for this paper directly from the 765 parent groups[1]. Every control group containing at least one CG_4 galaxy is excluded from the control samples, so no galaxy appears on both sides of any comparison; the extent of this compact-group/projected-core overlap is quantified in Appendix C. We also remove the systems associated with Lim group 3688, which appears to be spurious: its single outlying-redshift member ($z = 0.048$, against 0.032–0.034 for the other six) is selected into both our original $\mathrm{Control}_{4\mathrm{B}}$ and $\mathrm{Control}_{4\mathrm{C}}$ samples and inflates the group velocity dispersions to $\simeq 1980\,\mathrm{km\,s^{-1}}$. After these removals, the control samples contain 698 $\mathrm{Control}_{4\mathrm{B}}$, 704 $\mathrm{Control}_{4\mathrm{C}}$, and 56 RG_4 groups (2792, 2816, and 224 galaxies); the $\mathrm{Control}_{4\mathrm{B}}$ and $\mathrm{Control}_{4\mathrm{C}}$ counts differ because the luminosity-ranked and projected-distance-ranked core definitions select different galaxies and therefore retain different Lim groups after the compact-group exclusion.

We use SDSS DR16 (Ahumada et al. 2020) to retrieve stellar masses, star formation rates (SFRs; Brinchmann et al. 2004; Kauffmann et al. 2003b), and Galaxy Zoo morphologies (Lintott et al. 2008, 2011); the exact columns, table joins, and query are given in Appendix A. Throughout, 'SDSS selection' denotes the reference sample returned by that query: all spectroscopic galaxies with $0.005 \leq z \leq 0.0452$, extinction-corrected Petrosian $r \leq 17.77$, a valid MPA-JHU stellar mass, and Galaxy Zoo classifications. It serves as a field-dominated reference population, not as a control sample. All projected distances and derived quantities adopt the Planck 2015 cosmology (Planck Collaboration et al. 2016), consistently with our previous paper.

2.2. Star-formation and morphology classes

We classify galaxies from SDSS total sSFRs. In the MPA-JHU catalogue, values of −9999 in the sSFR fields are sentinel values indicating that no valid sSFR estimate is available, rather than physical measurements; the same missing-value convention is used for MPA-JHU DR8 target quantities by Zeraatgari et al. (2024). We therefore treat `specsfr_tot_p50 = −9999` as *missing data*: such galaxies are excluded from the star-formation classification and from every sSFR-based figure and fraction, and are reported as counts only (Table 2). All quenched and star-forming fractions quoted in this paper are computed among classified galaxies only. The missing-sSFR incidence is markedly higher in CG_4 (7% of members, and 10% of BGGs) than in the field selection (1%), plausibly reflecting crowded-field spectroscopy and deblending (Sect. 4.3) rather than a physical population. Galaxies with a measured sSFR are split into quenched and star-forming classes with a two-component Gaussian mixture model (GMM) in the $\log M_\star$–$\log \mathrm{sSFR}$ plane; BPT/AGN handling and the exact GMM boundary definition are given in Appendix B (Fig. B.1).

Morphologies are drawn from Galaxy Zoo 1 (Lintott et al. 2008, 2011), a single-question classification in which volunteers assigned each galaxy image to one of several morphological categories. The resulting debiased vote fractions are provided in the `zooSpec` table through the columns `p_el_debiased` (fraction voting "elliptical") and `p_cs_debiased` (fraction voting combined-spiral; Lintott et al. 2011). We classify a galaxy as elliptical if `p_el_debiased` exceeds 0.5, as spiral if `p_cs_debiased` exceeds 0.5, and as "uncertain" otherwise. These are vote-fraction classes, not a full structural decomposition. In particular, the catalogue does not provide a clean E/S0 separation, so references below to elliptical or early-type

[1] Our previous paper quotes 61 excluded projected-core groups; the archived parent catalogue reproducibly yields 60. The difference traces to an earlier revision of the parent catalogue and does not affect any comparison in this paper.

galaxies should be read as Galaxy Zoo 1 elliptical-vote proxies for early-type morphology rather than as a bulge-disc structural classification.

2.3. Galaxy sizes

Our primary galaxy-size measure is the circular half-light radius of the pure Sérsic fit in the r band, $R_{\rm chl,r}$, from the bulge+disk decomposition catalogue of Simard et al. (2011). These are PSF-convolved GIM2D fits with re-derived sky levels and deblending designed with crowded fields in mind – the relevant regime for compact-group photometry – so the fitted sizes are corrected for seeing. The catalogue is keyed to SDSS DR7 identifiers; we bridge our DR16 objects to their DR7 counterparts through the SkyServer `SpecDR7` cross-match table. Two guards protect this bridge: rows in which the Simard redshift differs from the catalogue redshift by more than 0.005 are treated as wrong matches and their structural values dropped (15 rows), and rows in which two distinct catalogue galaxies of the same group resolve to a single DR7 identifier are treated as blended detections and dropped (0 rows). The Simard sizes are tabulated in kpc under the catalogue's own cosmology; we convert them back to angular sizes with the catalogue plate scale and then to proper kpc with the same Planck 2015 conversion as every other projected quantity in this paper.

As a secondary size measure we use the SDSS Petrosian radii `petroR50_r` and `petroR90_r`, which also define the concentration index $C = R_{90,r}/R_{50,r}$. Petrosian radii are not corrected for seeing, and the SDSS Petrosian aperture misses a profile-dependent fraction of the flux, roughly 20% for an $n = 4$ profile, which biases $R_{50,r}$ low for concentrated galaxies (Blanton et al. 2001; Graham et al. 2005) – precisely the elliptical/early-type-like population in which CG_4 is enhanced. The Petrosian measure is nevertheless retained because it is model-independent, essentially fully available, and provides the like-for-like bridge to the compact-group size study of Coenda et al. (2012). All sizes are required to lie between 0.1 and 50 kpc after re-conversion; Sérsic fits pegged at the GIM2D bounds ($n_g \leq 0.55$ or $n_g \geq 7.9$; 63 galaxies) are excluded from the primary sample, and Petrosian entries must satisfy $R_{50,r} > 0$, $R_{90,r} > R_{50,r}$, and a finite positive uncertainty. The completeness of both measures, and its difference between CG_4 and the controls, is audited in Sect. 3.8.

2.4. Statistical design

Descriptive baseline comparisons use Fisher or Barnard exact tests for two-sample categorical tables, and rank-sum or bootstrap comparisons for distributional and median differences. Spearman and Pearson coefficients are used only as diagnostics of monotonic or linear trends; when the coefficients are small, we interpret them as weak even if the p-values are formally small.

The primary inference is organized around four complementary analyses (Table 1) that separate individual-galaxy, group-composition, and shared-host contrasts: (i) covariate-adjusted binomial models fit *separately* for each CG_4–control pair (Sect. 3.2); (ii) a one-to-one galaxy-level matched comparison; (iii) a group-level matched comparison of satellite composition; and (iv) a within-host stratified comparison of CG_4 members against the other members of the same host group. These analyses do not form a monotonic sequence of increasingly strong adjustment: they change the statistical unit, matched population, conditioning variables, outcome, effect scale, and interpretation. Throughout, the independence unit is the *physical* group: standard errors are clustered by Lim group (unifying the overlapping control definitions, with CG_4 systems assigned to their host Lim group), and all resampling is blocked at the group level, since the four galaxies of one group are not independent observations. Pooled models across the three controls are reported only as secondary summaries, after deduplicating the overlapping control samples to one row per physical galaxy (Appendix F). Bootstrap p-values use the add-one empirical convention $p = (k + 1)/(B + 1)$ with the number of resamples B reported, so no resampling p-value can fall below its Monte-Carlo resolution; displayed p-values are floored at 10^{-4}.

Table 1. Complementary inferential framework. Each analysis answers a different question about the CG_4 population; uncertainties are clustered or resampled by physical group.

Analysis	Unit	Estimand
Adjusted, per control	galaxy	individual conditional association at fixed galaxy and group covariates
Matched, galaxy	galaxy	matched individual-galaxy contrast on one-to-one pairs
Matched, group	group	difference in satellite composition between matched systems
Within host	member	CG_4 members versus non-CG_4 co-members of the same host

Holm or Benjamini–Hochberg corrections are applied within pre-defined families of related tests. These families are defined over the full set of computed diagnostics – including those reported only in the appendices – and are *not* recomputed over the subset of results shown in the main text. For the per-control group-level satellite-composition matches, the three control definitions are treated as separate scientific estimands because $\mathrm{Control}_{4B}$, $\mathrm{Control}_{4C}$, and RG_4 answer different control questions; their p-values are therefore not adjusted across control definitions, and the pooled match is a secondary summary rather than a fourth confirmatory comparison. The paired sign-flip permutation p-value is the primary paired test for these group-level matches; the bootstrap sign-crossing p-value and interval are used for robustness and effect-size uncertainty. A Holm sensitivity calculation for the three per-control bootstrap and permutation p-value sets is reported in Appendix F. Colours, AGN-like fractions, Hα equivalent widths, magnitude gaps, crossing-time trends, projected phase space, and the pairwise tidal index are secondary diagnostics: the conclusions below rely on signals that survive the adjusted, matched, and within-host analyses, not on isolated exploratory tests.

3. Results

3.1. Raw population contrasts

The sSFR status and Galaxy Zoo morphology mix of each sample are shown in Tables 2 and 3. For the morphologies, Barnard's two-sided exact tests give p-values of $p = 1.4 \times 10^{-3}$ for ellipticals and $p = 1.2 \times 10^{-3}$ for spirals ver-

Table 2. Number of galaxies in each sSFR class for each sample. Galaxies without a valid sSFR estimate ('No sSFR') are excluded from the classification: their percentages are quoted with respect to all galaxies of the (sub)sample, while the quenched and star-forming percentages are quoted with respect to classified galaxies only.

Sample	No sSFR	Quenched	Star-forming
CG_4	18 (7%)	146 (63%)	84 (37%)
BGG	*6 (10%)*	*45 (80%)*	*11 (20%)*
Satellites	*12 (6%)*	*101 (58%)*	*73 (42%)*
$Control_{4B}$	180 (6%)	1593 (61%)	1019 (39%)
BGG	*58 (8%)*	*533 (83%)*	*107 (17%)*
Satellites	*122 (6%)*	*1060 (54%)*	*912 (46%)*
$Control_{4C}$	179 (6%)	1551 (59%)	1086 (41%)
BGG	*59 (8%)*	*538 (83%)*	*107 (17%)*
Satellites	*120 (6%)*	*1013 (51%)*	*979 (49%)*
RG_4	5 (2%)	101 (46%)	118 (54%)
BGG	*3 (5%)*	*38 (72%)*	*15 (28%)*
Satellites	*2 (1%)*	*63 (38%)*	*103 (62%)*
SDSS selection	*634 (1%)*	*7705 (15%)*	*44192 (85%)*

Table 3. Number of galaxies in each Galaxy Zoo vote-fraction morphology proxy for each sample.

Sample	Elliptical	Spiral	Uncertain
CG_4	124 (50%)	86 (35%)	38 (15%)
$Control_{4B}$	1106 (40%)	1266 (45%)	420 (15%)
$Control_{4C}$	1157 (41%)	1140 (40%)	519 (18%)
RG_4	60 (27%)	129 (58%)	35 (16%)
SDSS selection	*10091 (19%)*	*34715 (66%)*	*7725 (15%)*

sus the $Control_{4B}$ sample, $p = 6.4 \times 10^{-3}$ and $p = 0.074$ versus $Control_{4C}$, and $p < 10^{-4}$ and $p < 10^{-4}$ versus RG_4. The galaxies in CG_4 systems, which we identified as frequent cores of richer groups in Tricottet et al. (2025), show a clear raw excess of elliptical classifications and a deficit of spiral classifications. Restricted to BGGs, CG_4 BGGs have a higher elliptical fraction (58.1%) than RG_4 BGGs (37.5%); the p-values obtained with two-sided Fisher exact test are 1.00 against $Control_{4B}$, 1.00 against $Control_{4C}$ and 0.03 against RG_4. Restricted instead to star-forming galaxies, the elliptical/spiral mix of CG_4 is statistically consistent with that of every control (Barnard two-sided exact test: $p = 0.283$ against $Control_{4B}$, $p = 0.964$ against $Control_{4C}$, and $p = 0.186$ against RG_4): we detect neither an excess nor a deficit of *star-forming* elliptical galaxies, so the raw morphology excess is not carried by star-forming ellipticals.

The star-formation contrast is weaker and rank-dependent. In the raw all-galaxy comparison, Fisher's exact test gives star-forming-fraction p-values of $p = 0.481$ versus the $Control_{4B}$ sample, $p = 0.184$ versus $Control_{4C}$, and $p = 3 \times 10^{-4}$ versus RG_4. For satellites, the corresponding p-values obtained with two-sided Fisher exact test are $p = 0.302$ against $Control_{4B}$, $p = 0.069$ against $Control_{4C}$ and $p = 2.3 \times 10^{-4}$ against RG_4; for BGGs, they are $p = 0.578$, $p = 0.576$, and $p = 0.37$ (same control order). The satellite deficit is clearest against RG_4. We treat all raw tests as descriptive, because star-formation class and morphology are entangled with stellar mass and luminosity; this motivates the adjusted analyses of Sect. 3.2.

For the star-forming subset, we fit the star-forming main sequence with polynomials of order 1–4, adopting order 2 as the lowest-order model with the smallest five-fold cross-validated RMS scatter (0.30523 dex; Appendix D). The median CG_4-minus-control residual offsets about this relation are small, control-dependent, and change in strength after matching (Appendix D), so we treat main-sequence residuals as a secondary diagnostic rather than as evidence for a robust compact-group star-formation offset.

3.2. Per-control adjusted contrasts (primary)

The three control samples answer different questions, so our primary galaxy-level inference fits CG_4 against each control *separately*: against $Control_{4B}$ (is the CG_4 population special compared with the luminous population of richer ordinary groups?), against $Control_{4C}$ (compared with BGG-centred projected cores?), and against RG_4 (compared with true four-member ordinary groups?). Each contrast fits the same family of adjusted binomial models (stellar mass, redshift, rank, group luminosity, and velocity dispersion where sufficiently complete), with standard errors clustered by *physical* group: the Lim group for controls, and for CG_4 systems their host Lim group where one exists, so that the same physical system never enters as more than one cluster. Holm correction is applied within each contrast across its model family.

Against $Control_{4B}$, the elliptical odds ratio is 2.14 (95% CI $[1.58, 2.90]$; Holm-corrected $p < 10^{-4}$), and the quenched odds ratio is 1.80 (Holm-corrected $p = 0.003$).

Against $Control_{4C}$, the elliptical odds ratio is 1.30 (95% CI $[0.98, 1.73]$; Holm-corrected $p = 0.443$), and the quenched odds ratio is 0.91 (Holm-corrected $p = 1.000$).

Against RG_4, the elliptical odds ratio is 2.81 (95% CI $[1.79, 4.38]$; Holm-corrected $p < 10^{-4}$), and the quenched odds ratio is 1.40 (Holm-corrected $p = 0.476$).

The asymmetry between the contrasts is itself informative. The adjusted individual-galaxy morphology excess is strong against the luminosity-selected cores of richer groups and against true four-member groups, but is not detected against the BGG-centred *projected* cores of ordinary groups, the control definition that most closely mimics a compact projected configuration. Evidence against $Control_{4C}$ therefore depends on the estimand: the adjusted individual-galaxy contrast is not detected here, whereas the group-level satellite-composition contrast below is positive. This pattern anticipates the local projected-density interpretation of Sect. 3.6 (Fig. 1).

As a compact secondary summary, a pooled model over the deduplicated control catalogues (one row per physical galaxy, cluster-robust by physical Lim group; full family in Appendix F) gives the same qualitative picture: the pooled elliptical odds ratio is 1.50 (Holm-corrected $p = 0.032$), and the pooled quenched odds ratio is 1.08 (Holm-corrected $p = 1.000$). The adjusted morphology signal is primarily carried by satellites rather than BGGs: the satellite-only models give an elliptical odds ratio of 1.71 and a spiral odds ratio of 0.59, whereas the BGG elliptical coefficient is not detected after adjustment (OR 0.77; Holm-corrected $p = 1.000$) (Fig. 2).

3.3. Matched comparisons

Group-level matched satellite-composition contrast (primary). Because the four galaxies of a compact group are not independent, the group-level matched estimand treats each

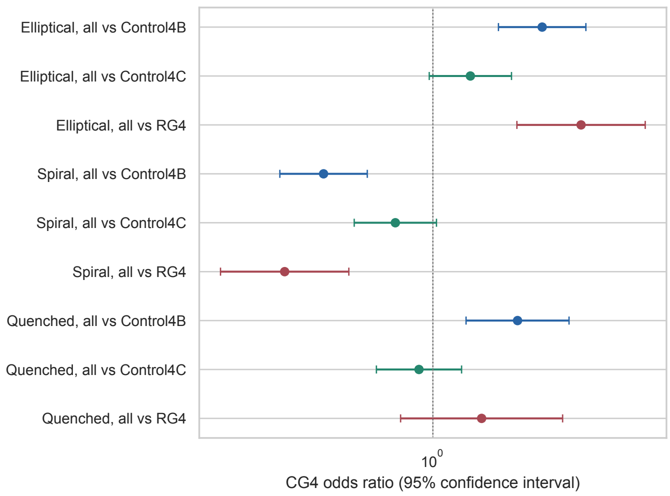

Fig. 1. Adjusted CG_4 odds ratios from the three separate per-control contrasts (colour-coded by control sample) for the all-member elliptical, spiral and quenched outcomes. Bars show 95% confidence intervals; the vertical line marks equal odds.

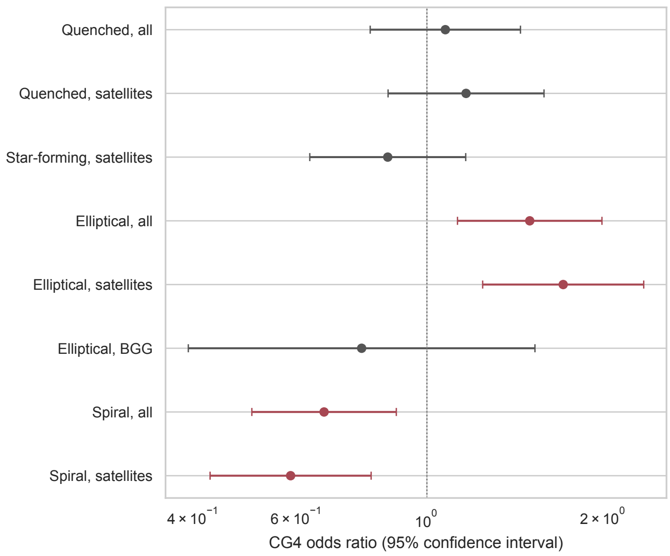

Fig. 2. Adjusted CG_4 odds ratios for star-formation and morphology diagnostics from the pooled secondary models, including all-galaxy, satellite-only, and BGG-only morphology rows. Bars show 95% confidence intervals; the vertical line marks equal odds.

quartet as the statistical unit: for each group we count its elliptical-vote satellites ($N_{\rm ell,sat} \in \{0, 1, 2, 3\}$) and match CG_4 groups one-to-one to control groups on redshift, BGG stellar mass, total quartet luminosity, and velocity dispersion (propensity caliper 0.2 SD of the logit). We run the matching three times with the pool restricted to a single control type (one quartet per physical Lim group within each type), and once with the pool drawn from all three types combined. For the per-control matches, the paired sign-flip permutation p-value is the primary p-value; the bootstrap sign-crossing probability and bootstrap interval quantify robustness and effect-size uncertainty.

The pooled control match covers 54 CG_4 groups and 634 control groups, yielding 54 matched pairs, drawn predominantly from Control_{4B} (47 Control4B, 5 RG4, 2 Control4C, reflecting the pool composition after deduplication). This pooled match is a secondary summary because the control catalogues overlap. Its elliptical-vote satellite fraction difference is $\Delta = 0.160$ (95% bootstrap interval $[0.019, 0.299]$; bootstrap $p = 0.027$, permutation $p = 0.031$; $B = 9999$).

The per-control matches (Table 4) give a consistent picture under the stated multiplicity convention: the CG_4-minus-control fraction difference is positive and nominally detected against all three control types. Control_{4C} is the physically most stringent comparison, since its projected-core definition most closely mirrors the CG_4 compactness criterion; against Control_{4C}, the match yields 53 pairs with $\Delta = 0.148$ (95% CI $[0.025, 0.267]$; primary permutation $p = 0.027$, bootstrap $p = 0.021$), so the satellite-composition contrast remains nominally detected even against projected ordinary-group cores. The galaxy-level match below estimates a different individual-galaxy contrast rather than a weaker version of this group-composition result.

Table 4. Per-control group-level matched satellite-composition contrasts. Δ is the matched mean difference in elliptical-vote satellite fraction (CG_4 minus control). The paired sign-flip permutation p-value is primary; the bootstrap p-value is the add-one sign-crossing probability used as a robustness check, with $B = 9999$ resamples. P-values are not adjusted across the three control definitions; Appendix F reports the Holm sensitivity. The pooled row is secondary.

Control	Pairs	Δ	95% CI	$p_{\rm perm}$ / $p_{\rm boot}$
Control_{4B}	54	0.160	$[0.034, 0.284]$	0.016 / 0.012
Control_{4C}	53	0.148	$[0.025, 0.267]$	0.027 / 0.021
RG_4	28	0.196	$[0.036, 0.351]$	0.026 / 0.019
Pooled	54	0.160	$[0.019, 0.299]$	0.031 / 0.027

Galaxy-level matched contrast (secondary). We matched 234 of the 248 available CG_4 galaxies one-to-one to 234 ordinary-group galaxies without replacement, requiring common propensity-score support, a 0.2-standard-deviation caliper and exact rank strata; the control pool is first deduplicated by SDSS objid, no control may share an objid with a CG_4 galaxy, and no control galaxy is used more than once. The propensity model uses $\log M_\star$, redshift, rank, $\log L_{\rm group}$, and σ_v; projected distance is retained as a phase-space and interaction diagnostic rather than matched away, because it partly defines compactness and has limited overlap. The largest absolute standardized mean difference changed from 0.24 before matching to 0.11 afterwards; the residual imbalance is carried by $\log L_{\rm group}$ and lies slightly above the conventional 0.1 threshold, which is why we read the matched contrasts together with the adjusted models rather than in isolation. Matched differences are quoted with group-blocked bootstrap intervals: pairs are resampled by the treated galaxy's compact group (never galaxy by galaxy), with $B = 9999$ resamples and add-one empirical p-values. Pool composition, provenance, and balance details are given in Appendix F.

The matched CG_4-minus-control difference in elliptical fraction is 0.087, with 95% bootstrap interval $[-0.005, 0.179]$ and Holm-corrected $p = 0.361$: directionally consistent with the adjusted models, but not detected as an individual-galaxy matched contrast once the four galaxies of a group are resampled together. The matched quenched-fraction difference is -0.022 ($[-0.100, 0.055]$;

Holm-corrected $p = 1.000$), so the matched data do not establish an enhanced quenched fraction after correction; the star-forming and spiral fractions are exact complements of these outcomes on their complete-case subsets and are reported in Appendix F. Among matched star-forming galaxies, the residual-sSFR difference is 0.074 ($[-0.075, 0.226]$; $p = 0.332$).

The matched-outcome Holm correction is intentionally conservative: quenched/star-forming and elliptical/spiral diagnostics are retained in the same family even though they are exact complements on their complete-case subsets. Using a non-redundant family would not weaken the morphology conclusion.

The matched individual-galaxy comparison is therefore compatible with the adjusted models but less decisive for morphology than the group-composition estimand; the quenched and star-forming-fraction contrasts are more model-dependent.

3.4. Compact-group members versus other members of the same host group

For the Embedded and Predominant CG_4 systems, whose quartets live inside richer Lim parent groups, the sharpest available test of "compact subconfiguration versus shared host environment" compares the CG_4 member galaxies with the *non-CG_4 members of the same host group*. The primary estimator is conditional logistic regression stratified by host group: the stratum-level intercept is conditioned out in the partial likelihood, so every host-level confounder shared by all members of the same host (halo mass, richness, redshift, large-scale environment) is removed by design. As a robustness check we also fit a binomial GLM with host-group fixed-effect dummies and cluster-robust standard errors by host (FE-GLM). The design starts from 54 host groups (958 member galaxies, of which 224 belong to a CG_4); the models include stellar mass, host luminosity rank, and projected host-centric radius as galaxy-level covariates.

For the elliptical-vote outcome, the complete-case conditional-logit sample contains 791 galaxies in 53 host strata; the remaining design host has no usable Galaxy Zoo elliptical/spiral classification after the model complete-case filters. All 53 complete-case host strata are informative; no concordant strata are dropped. The elliptical-vote odds ratio for CG_4 membership is 1.47 (95% CI $[0.92, 2.35]$; Holm-corrected $p = 0.216$), and the quenched odds ratio is 0.75 (Holm-corrected $p = 0.269$; 915 complete cases in 54 host strata). The FE-GLM yields a similar elliptical-vote point estimate but a smaller estimated uncertainty and nominal significance: OR 1.57 (95% CI $[1.02, 2.43]$; nominal $p = 0.041$). We retain the conditional-logit estimator as primary because it conditions out the host-specific intercept directly.

At fixed stellar mass, rank, and host-centric radius, no residual excess is detected for compact-subconfiguration membership *within* a shared host under the primary conditional estimator. The confidence interval nevertheless permits a range of moderate effects, so this result does not establish equality. It is consistent with the per-control asymmetry of Sect. 3.2: the CG_4 excess appears more clearly against galaxies of *other* groups than against co-members of the same host, pointing to dense projected-core location rather than compact-group membership alone.

Table 5. Robustness of the CG_4 morphology contrast. Odds ratios (OR) are adjusted CG_4 coefficients; the continuous row is the adjusted offset in the debiased elliptical vote fraction p_E. P-values in this compact diagnostic table are unadjusted unless explicitly labelled.

Test	Result
Exclude neighbours within 55 arcsec	elliptical fraction 52.5% (CG_4) vs 41.1% ($\mathrm{Control}_{4C}$); $p = 0.002$
Close-neighbour-flag model	OR 1.52 $[1.13, 2.04]$; $p = 0.006$
Threshold sweep 0.4–0.8	OR 1.46–1.98
Continuous p_E offset	0.070 $[0.027, 0.112]$; $p = 0.001$
Sérsic $n_g > 2.5$ cross-check	OR 1.42; $p = 0.018$

3.5. Morphology robustness

Because Galaxy Zoo morphology is image-based and the adopted vote-fraction cut is one of several defensible choices, we subject the morphology contrast to a battery of robustness tests, summarized in Table 5.

After excluding galaxies whose nearest projected neighbour within the catalogue group lies within 55 arcsec (retaining 200 CG_4 and 2455 $\mathrm{Control}_{4C}$ galaxies), the CG_4 elliptical fraction *rises* relative to the raw full-sample value in Table 3, rather than falling, and the CG_4–$\mathrm{Control}_{4C}$ contrast remains nominally significant in this diagnostic. This argues against the simplest interpretation that the elliptical excess is produced solely by crowding in the closest pairs. The close-neighbour flag is itself associated with elliptical-vote morphology (OR 2.07; $p < 10^{-4}$), so this test does not prove that Galaxy Zoo classifications are immune to crowding; it rules out the closest-pair artefact as the sole origin of the result.

Across classification thresholds of 0.4–0.8 on the deduplicated pooled sample, the adjusted CG_4 elliptical odds ratio varies only mildly and never changes sign (Table 5; the unclassified fraction grows from 13% to 50% over the sweep). A threshold-free check models the debiased elliptical vote fraction p_E itself, and a structural cross-check independent of Galaxy Zoo uses the Simard et al. (2011) pure-Sérsic index (early type: $n_g > 2.5$; 77% coverage); both confirm the direction of the contrast, the latter subject to the differential Simard completeness discussed in Sect. 3.8.

3.6. Pairwise tidal index

The pairwise analysis reconstructs projected separations and velocity differences for 6080 galaxies and refits the adjusted models with and without the pairwise tidal index $\sum_j M_{\star,j}/R_{ij}^3$ (in $\mathrm{M}_\odot\,\mathrm{kpc}^{-3}$). The two model variants estimate *different quantities*. The model without the index estimates the *total* association between compact-group membership and the outcome at fixed stellar mass, redshift, rank, and group-scale covariates; the model with the index estimates the association *at fixed projected local density*. Because the index is built from R^{-3}-weighted projected separations, it is mathematically entangled with the compact-group selection itself (compact groups are selected to have small projected separations), so conditioning on it partially conditions on the exposure. A reduction of the CG_4 coefficient when the index is added is thus *conditional attenuation*, not mediation: it does not show that local density accounts for the morphology contrast, and it cannot, because the causal ordering of selection, density, and morphology is not identified by projected snapshots. The baseline models in this subsection are refit on the pairwise-analysis complete-case sample, so their coefficients need not be numerically identical to the corresponding coefficients in Sect. 3.2 or Appendix F; because R_{ij} is projected, the term is a local projected-density and immediate-neighbour proxy rather than a direct tidal-history measurement.

For elliptical morphology, the baseline CG_4 odds ratio is 1.64 ($p = 4.4 \times 10^{-4}$), whereas the model including the tidal index gives a CG_4 odds ratio of 1.13 ($p = 0.421$); the tidal-index term itself has odds ratio 1.58 ($p < 10^{-4}$). The projected tidal-density proxy therefore *attenuates* the CG_4 coefficient; given the selection entanglement described above, this conditional attenuation must not be read as the index accounting for the effect, and the result remains an exploratory projected-pair diagnostic.

For quenched status, the baseline CG_4 odds ratio is 1.17 ($p = 0.291$), while the model including the tidal index gives 0.77 ($p = 0.106$), with a tidal-index term of odds ratio 1.68 ($p < 10^{-4}$).

3.7. Secondary outcomes

Star formation. The adjusted and matched analyses above already bound the star-formation contrast: the per-control quenched odds ratios are control-dependent (Sect. 3.2), the matched quenched and star-forming fractions show no detected offset after correction (Sect. 3.3), and the main-sequence residuals are a secondary diagnostic (Sect. 3.1).

The interpretation does not rely on the exact GMM division of the sSFR plane: repeating the key satellite comparison with a fixed threshold, $\log_{10}(\mathrm{sSFR/yr^{-1}}) = -11.0$, gives CG_4 and RG_4 satellite star-forming fractions of 41.4% and 60.2%, respectively, a difference of −18.8 percentage points ($p = 5.3 \times 10^{-4}$). The qualitative hierarchy is unchanged: the morphology result is independent of the sSFR threshold, whereas the quenched/star-forming contrast remains more sensitive to modelling choices.

Hα emission. Our retrieved catalogues do not include the age-sensitive indices D_n4000 and $\mathrm{H}\delta_A$ (available in the MPA-JHU `galSpecIndx` product), so we defer a post-starburst classification to follow-up work and perform only a limited Hα-equivalent-width comparison, using the SDSS convention in which negative equivalent width denotes emission and classifying strong Hα emission by $\mathrm{H}\alpha\,\mathrm{EW} \le -3$ Å.

The strong-emission fraction is 32.3% in CG_4, 41.0% in $\mathrm{Control}_{4\mathrm{B}}$, 44.2% in $\mathrm{Control}_{4\mathrm{C}}$, 56.9% in RG_4.

Relative to $\mathrm{Control}_{4\mathrm{B}}$, the CG_4 strong-emission fraction differs by −0.087 (Holm-corrected $p = 0.012$).

Relative to $\mathrm{Control}_{4\mathrm{C}}$, the CG_4 strong-emission fraction differs by −0.119 (Holm-corrected $p = 0.001$).

Relative to RG_4, the CG_4 strong-emission fraction differs by −0.246 (Holm-corrected $p < 10^{-4}$).

These limited results indicate a lower strong-emission fraction in CG_4 than in each control, but they do not provide a direct post-starburst or recently-quenched classification.

Emission-line and AGN-like populations. The emission-line analysis classifies BPT star-forming, composite, and AGN-like populations for galaxies with the required diagnostic line ratios: composite galaxies lie between the Kauffmann et al. (2003a) and Kewley et al. (2001) demarcations in the [N II] BPT plane, and AGN-like galaxies lie above the Kewley et al. (2001) line; galaxies with a pre-existing AGN flag but no BPT line ratios are also counted as AGN-like, and BPT-unclassified galaxies are excluded from the AGN-like fraction denominator rather than treated as confirmed non-AGN. Among BPT-classified CG_4 galaxies, the AGN-like fraction is 38.0%.

After mass and environment adjustment, the CG_4 AGN-like odds ratio is 1.14 ($p = 0.447$): AGN-like activity does not provide a robust independent compact-group discriminator.

Magnitude gaps and fossil-like assembly. We define $\Delta m_{12} = M_{r,2} - M_{r,1}$ and $\Delta m_{14} = M_{r,4} - M_{r,1}$, so that a more dominant BGG has a positive, larger gap.

For Δm_{12}, the CG_4 and control medians are 1.17 and 1.10 mag, respectively (Holm-corrected $p = 0.824$); for Δm_{14}, they are 2.56 and 2.59 mag (Holm-corrected $p = 0.824$).

The compact-group morphology signal is therefore not accompanied by a detected excess in these simple fossil-like magnitude-gap indicators.

The correlation between Δm_{12} and quenched satellite fraction is $\rho_S = -0.14$ (Holm-corrected $p < 10^{-4}$).

Projected phase space. We ask whether the CG_4–RG_4 satellite contrasts vary with projected position relative to the BGG. The analysis uses 174 CG_4 and 166 RG_4 satellites, excludes rank-1 galaxies, ranks satellites by projected BGG-centric distance, and, when available, uses $|\Delta v_{\mathrm{BGG}}|/\sigma_v$ as a projected phase-space diagnostic; such coordinates are statistical proxies for infall state or quenching time, not orbital reconstructions (Mahajan et al. 2011; Oman & Hudson 2016; Wetzel et al. 2013).

In the innermost satellite-distance bin, the quenched fractions are 68.3% for CG_4 and 48.2% for RG_4, a difference of 20.1 percentage points; the corresponding early-type fractions are 64.7% and 35.6% (difference 29.1 points).

The quenched-fraction contrast is similar between the inner and outer distance ranks, so the data do not isolate the CG_4 offset to the immediate BGG environment.

After adjusting for stellar mass, redshift, distance bin, and available group-scale controls, the quenched-status model gives a CG_4 odds ratio of 1.18 (95% CI 0.43–3.24; $p = 0.742$), and the analogous early-type model gives 2.42 ($p = 0.078$).

A nearest-neighbour mass–redshift matched check on 126 CG_4–RG_4 satellite pairs gives matched quenched- and early-type-fraction differences of 9.5 and 18.7 percentage points, respectively.

In compact groups the member counts are small, velocity dispersions are noisy, and projection mixes recently accreted and long-resident satellites, so we treat these bins as an observational diagnostic of where the contrast appears, not as a dynamical clock.

Optical colours. We derive the $(u-r)$, $(u-g)$, $(g-r)$ and $(r-i)$ colours from the matched SDSS photometry and, because stellar mass confounds colour comparisons (Alpaslan et al. 2015), model each colour jointly with $\log(M_\star/M_\odot)$ and redshift, with cluster-robust standard errors. The colour-analysis subset is not representative of the full galaxy catalogues – matched galaxies are systematically less massive, more actively star-forming, and more frequently satellites than unmatched galaxies – so colours are treated as a secondary diagnostic. In the pooled satellite comparison, no blue CG_4 coefficient is detected after Holm correction, and the adjusted coefficients are control-sample dependent. The data therefore do not establish a robust global blue or red offset for CG_4 satellites; the full colour–mass, model-dependence, and radial diagnostics are given in Appendix E.6.

Group-scale diagnostics. A broader set of environmental diagnostics is reported in Appendix E: significant monotonic sSFR–distance trends in individual samples (Appendix E.1); dominance splits whose clearest separation is in group luminosity rather than in any CG_4-specific property; morphology tests against the Zheng–Shen class, Δm_{12}, and the BGG luminosity fraction, none of which survives false-discovery-rate control; and a crossing-time diagnostic that primarily tracks virial mass and mass-to-light ratio.

3.8. Galaxy sizes at fixed stellar mass

We now test whether CG_4 galaxies differ in size at fixed stellar mass, using the seeing-corrected Simard half-light radii as the primary measure and the Petrosian radii as the model-independent cross-check (Sect. 2.3). The Simard measure resolves 77.4% of the CG_4 galaxies and 72.2% of the pooled control galaxies, while the Petrosian radius is essentially complete in every sample (Appendix Fig. G.1). Because the CG_4–control difference in Simard completeness exceeds five percentage points, every Simard-based coefficient below carries a differential-completeness caveat and is read together with the Petrosian rerun, which does not share this incompleteness.

For satellite galaxy size, the CG_4 offset at fixed stellar mass, redshift, and available group-scale covariates is 0.009 dex (2.0%), with 95% confidence interval $[-0.021, 0.039]$ and Holm-corrected $p = 1.000$, over 2928 satellites in 757 physical groups.

The all-galaxy model gives 0.010 dex (95% CI $[-0.017, 0.037]$; Holm-corrected $p = 1.000$), and the BGG-only model gives 0.019 dex (Holm-corrected $p = 1.000$).

No coefficient in this primary family is detected after Holm correction, so the pooled adjusted models do not establish a CG_4 size offset at fixed stellar mass.

The per-control versions of the fit are descriptive (Appendix Fig. G.2); the fitted CG_4 coefficient is -0.035 dex against $\mathrm{Control}_{4\mathrm{B}}$, 0.027 dex against $\mathrm{Control}_{4\mathrm{C}}$, -0.006 dex against RG_4 (raw p-values with Benjamini–Hochberg control inside this figure family). The sign and significance of the CG_4 coefficient depend on the adopted control, so we do not interpret any single per-control contrast on its own.

The one-to-one matched comparison re-derives the same 234 propensity pairs as the matched-control analysis, with a separate Holm family for the three size outcomes so that the published matched-outcome family is untouched. On the 160 pairs in which both members have valid Simard sizes, the mean CG_4-minus-control difference in $\log_{10} R_{\mathrm{chl,r}}$ is 0.027 dex (95% bootstrap interval $[-0.013, 0.065]$; Holm-corrected $p = 0.556$). The corresponding Petrosian difference is 0.005 dex (Holm-corrected $p = 1.000$), and the matched concentration difference is 0.005 (Holm-corrected $p = 1.000$).

The Petrosian rerun, with the field seeing as an additional standardized covariate, gives a satellite coefficient of -0.004 dex (Holm-corrected $p = 1.000$) and an all-galaxy coefficient of 0.001 dex (Holm-corrected $p = 1.000$).

The concentration index $C = R_{90,r}/R_{50,r}$ provides a model-independent structural proxy: the adjusted satellite coefficient is 0.057 (Holm-corrected $p = 0.599$), so no CG_4 concentration-index shift is detected at fixed mass.

Crowding, measure-difference, tidal-index, radial, and morphology-stratified size diagnostics are given in Appendix G.

4. Discussion: drivers of structural evolution in compact groups

4.1. The morphology hierarchy and its estimands

The clearest residual difference between CG_4 and the regular-group controls is morphological: a higher Galaxy Zoo 1 elliptical-vote fraction and a lower spiral-vote fraction, carried primarily by satellites. The inferential framework of Sect. 3 shows, however, that the strength of this contrast depends on the estimand. At the galaxy level, the per-control adjusted excess is strong against $\mathrm{Control}_{4\mathrm{B}}$ and RG_4 but not detected against the projected cores $\mathrm{Control}_{4\mathrm{C}}$. At the group level, the matched elliptical-vote satellite fraction is higher in CG_4 than in every control (Table 4), including $\mathrm{Control}_{4\mathrm{C}}$ under the stated per-control convention. The galaxy-level matched contrast is directionally consistent but not detected once the four galaxies of a group are resampled together, and the primary within-host stratified estimator of Sect. 3.4 detects no additional residual excess of CG_4 members over co-members of the same host.

These results are not contradictory because the analyses differ in their matched populations, conditioning sets, outcomes, and effect scales. Since each quartet contributes three satellites, the distinction is not simply one of weighting galaxies versus groups. The group-level analysis matches systems on group properties and estimates a difference in satellite composition, whereas the galaxy-level analyses estimate individual conditional odds ratios or matched individual-galaxy differences after adjustment for galaxy- and group-level covariates. The conditional odds-ratio scale is also non-collapsible, and the uncertainty structures differ between clustered model-based inference, paired group resampling, paired sign-flip tests, and within-host conditioning. The observed pattern is therefore best read as estimand dependence: the CG_4 satellite-composition contrast remains visible at the system level, while the adjusted individual-galaxy and shared-host contrasts are weaker or not detected under their own conditioning sets.

The pairwise tidal-index experiment strengthens this interpretation while also narrowing it. For elliptical morphology, the pairwise complete-case baseline CG_4 odds ratio is 1.64 ($p = 4.4 \times 10^{-4}$). After adding the projected pairwise tidal-index proxy, the CG_4 odds ratio decreases to 1.13 ($p = 0.421$), while the tidal-index term itself has odds ratio 1.58 ($p < 10^{-4}$). This behaviour is expected if compact groups are selecting galaxies in locally dense projected configurations where repeated close companions, rather than the catalogue label alone, are linked to the elliptical/early-type-like population.

This proxy uses projected separations, so it may be inflated by line-of-sight alignments and is partly entangled with compact-group selection itself: compact groups are selected to be compact in projection. We therefore read the tidal-index result as a continuous local projected-density or immediate-neighbour proxy, not as a direct time-dependent tidal-field measurement. The $\mathrm{Control}_{4C}$ comparison makes the test deliberately stringent, because $\mathrm{Control}_{4C}$ is also selected as a projected compact core around a BGG. The persistence of a CG_4–$\mathrm{Control}_{4C}$ morphology contrast in the closest-neighbour exclusion test shows that the CG_4 label is not reduced to the simplest crowding artefact; at the same time, the tidal-index model indicates that the CG_4 coefficient attenuates after conditioning on a more continuous projected-density proxy. We therefore interpret the residual CG_4–$\mathrm{Control}_{4C}$ contrast as consistent with CG_4 selecting a more extreme part of the local-density distribution, rather than as proof that compact groups define a separate causal channel.

This reading is consistent with compact groups identifying the high-density, high-interaction tail of group environments – some genuinely compact and dynamically evolved, others dense cores or substructures inside larger haloes – with the physically relevant quantity being the accumulated local interaction history, whether in the present configuration or through preprocessing. It also matches the construction result that many compact groups are embedded in or associated with larger structures, and the stronger structural signatures reported for non-isolated compact groups in recent S-PLUS work. We note that no harmonised halo-mass estimate is available for these catalogues (Appendix F.4), so, although the adjusted models control for group luminosity and velocity dispersion, a residual halo-mass contribution to the morphology signal cannot be fully excluded.

4.2. Elliptical-vote morphologies, S0 degeneracy, and transition galaxies

The Galaxy Zoo morphology variables used here identify galaxies with high elliptical vote fractions and galaxies with high combined-spiral vote fractions. They do not cleanly distinguish classical ellipticals from S0 galaxies, faded disks, or other bulge-dominated disk systems. The enhanced CG_4 elliptical fraction should therefore be interpreted as a shift toward early-type-like appearances, not as proof that compact groups specifically build classical elliptical galaxies.

This caveat matters when comparing with the recent S-PLUS structural studies of compact-group galaxies by Montaguth et al. (2023, 2025, 2026a,b), which identify transition galaxies and trends in the R_e–Sérsic-index and mass-size planes, with especially strong signals in non-isolated compact groups and host structures. Their transition population can plausibly overlap with the elliptical-vote class in our catalogue: systems with faded disks, increasing central concentration, or S0-like structure may receive high elliptical vote fractions here even if they are not classical ellipticals. The two approaches are therefore complementary: their structural measurements describe the likely physical form of the transformation, while our regular-group controls show that the strongest residual compact-group signal is the accumulated elliptical/spiral vote-fraction balance. With the Simard structural parameters now attached to this sample, the R_e–Sérsic-n plane is measured directly for CG_4 and its controls (Appendix Fig. G.3), turning this comparison from a qualitative statement into a direct one.

4.3. Why the star-formation and size signals are weaker

The star-formation diagnostics do not follow the morphology signal one-to-one. CG_4 satellites have a lower raw star-forming fraction, especially relative to RG_4, but the adjusted evidence is control-dependent (detected only against $\mathrm{Control}_{4B}$) and the matched contrasts provide no evidence for a quenched-fraction difference. Among galaxies that remain star-forming, the main-sequence residuals do not show a robust compact-group offset. The colour analysis is control-sensitive and based on a biased colour-matched subset, while AGN-like fractions, magnitude gaps, and the limited Hα diagnostic do not identify a simple alternative driver.

The star-formation non-detections are also limited by the spectroscopy. CG_4 has a higher fraction of galaxies with a nearest projected neighbour below the 55-arcsec SDSS fibre-collision scale than the controls (19.4% versus 9.2%). The control value is pooled over the regular-control comparison; the $\mathrm{Control}_{4C}$-only exclusion fraction in Sect. 3.5 is larger (12.8%) because $\mathrm{Control}_{4C}$ is selected as the compact-core analogue. Spectra are therefore preferentially incomplete in the closest projected pairs, and short-lived interaction-triggered starbursts or post-starburst phases could be underrepresented. The MPA-JHU aperture corrections use photometric information outside the fibre (Brinchmann et al. 2004; Kauffmann et al. 2003a); these corrections may be less secure for interacting or blended compact-group members. We therefore treat the sSFR, SFMS, and Hα results as conservative secondary non-detections, not as proof that compact groups have no star-formation effect. This decoupling suggests that the dominant observable difference is not an instantaneous quenching event caught at the present epoch: a galaxy can lose visible spiral structure, build a more bulge-dominated profile, or fade into an S0-like state on timescales that do not require a simultaneously large sSFR residual in the surviving star-forming subset.

The size test carries a parallel message. Coenda et al. (2012) compared galaxies in the McConnachie et al. (2009) compact groups with loose-group and field samples using SDSS Petrosian half-light radii and found compact-group galaxies to be smaller at fixed luminosity, with the clearest offset for early types relative to the field, based on approximately 1σ comparisons of fitted size–luminosity relations. The present analysis re-poses that question with the non-split HMCG-based CG_4 sample against three regular-group controls, seeing-corrected model sizes, a Petrosian cross-check, cluster-robust adjusted models, one-to-one matching, crowding tests, and Holm-corrected confidence intervals.

Under this stricter design, we find no compact-group size offset at fixed stellar mass relative to regular groups. This does not contradict Coenda et al. (2012) directly, because their comparison was against loose groups and the field, whereas ours is against regular-group cores selected from the same parent catalogues.

Two interpretation guardrails apply: a negative Δ at fixed mass can reflect genuine tidal truncation or stripping, but also a higher bulge dominance in the CG_4 population or residual measurement blending in crowded fields. The morphology-stratified fits address the former, and the Petrosian cross-check, measure-difference diagnostic, and crowding exclusions bound the latter (Appendix G); any future claim of compact-group size evolution should be read through those blocks together rather than through a single pooled coefficient.

5. Conclusion

We have tested whether galaxies in CG_4 compact groups remain special after removing split systems, comparing them with three regular-group analogues drawn from the same parent catalogues – RG_4, $Control_{4B}$, and $Control_{4C}$ – through complementary individual-galaxy, group-composition, and shared-host contrasts (Table 1).

The main result is morphological, and it depends on the estimand. CG_4 galaxies – primarily satellites – have a higher Galaxy Zoo 1 elliptical-vote fraction and a lower spiral-vote fraction. The adjusted individual-galaxy excess is strong against the luminous cores of richer ordinary groups ($Control_{4B}$) and against true four-member groups (RG_4), but is not detected against BGG-centred projected cores ($Control_{4C}$). At the group level, matched CG_4 systems show positive elliptical-vote satellite-composition contrasts against all three controls under the stated per-control multiplicity convention (Table 4), including $Control_{4C}$. Under the primary conditional within-host estimator, no additional residual excess is detected for CG_4 members relative to non-CG_4 co-members of the same host. The morphology result is stable to the Galaxy Zoo 1 vote threshold, is recovered with a threshold-free vote-fraction model and a Sérsic-index cross-check, and persists after excluding the closest projected pairs; BGG morphology is not detected as different after adjustment. Conditioning on a local projected-density proxy attenuates the galaxy-level coefficient, but this conditional attenuation does not establish mediation.

The star-formation result is weaker. CG_4 satellites have a lower raw star-forming fraction, especially relative to RG_4, but the adjusted evidence is control-dependent, the matched contrasts provide no evidence for a quenched-fraction difference, and star-forming CG_4 galaxies show no robust offset from the fitted star-forming main sequence.

At fixed stellar mass, redshift, and group-scale covariates, no CG_4 offset is detected in seeing-corrected half-light radius, in either the adjusted models or the matched pairs.

The remaining secondary diagnostics do not identify a simple alternative driver: the colour results are control-sensitive and based on a biased matched subset, AGN-like fractions and magnitude gaps are not detected as parallel compact-group offsets, the Hα diagnostic is limited by the absence of age-sensitive spectral indices from our retrieved catalogues, and projected phase-space trends do not localize the contrast uniquely to the innermost satellites.

We therefore interpret CG_4 systems as hosting a population shifted towards early-type-like morphologies, consistent with accumulated structural transformation in dense group cores and/or preprocessing. This is not proof of a uniquely isolated present-day compact-group quenching mechanism. Explicit E–S0–spiral and transition-galaxy classifications, bulge–disc structure, homogeneous halo masses, and age-sensitive stellar-population diagnostics will be needed to determine whether the elliptical-vote satellite excess reflects disk fading, tidal heating, or merger-driven transformation in dense group cores.

Data availability

The input data are drawn from public SDSS spectroscopic and photometric tables, the MPA-JHU value-added catalogues, Galaxy Zoo classifications, the Simard et al. (2011) structural catalogue (VizieR `J/ApJS/196/11`), and the public group and compact-group catalogues cited in Sect. 2. The derived galaxy and group tables from this analysis will be made available through the CDS at publication. The full analysis and paper-generation code is available at https://github.com/MatthieuTricottet/CG_Gals; the release corresponding to this article will be archived with a DOI (e.g. Zenodo) at acceptance. Supplementary diagnostic figures removed from the printed appendices (main-sequence, environmental, phase-space, colour, and size diagnostics) are included in the archived release.

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Appendix A: SDSS data query

We specifically extract the columns `sfr_tot_p50`, `specsfr_tot_p50`, and `lgm_tot_p50` from the `galSpecExtra` table, and `p_el_debiased` and `p_cs_debiased` from the `zooSpec` table. In the query, `galSpecExtra`, `galSpecLine`, and `zooSpec` are joined to `SpecObj` by `specObjID`, while the photometric object is linked through `SpecObj.bestObjID=PhotoObj.objID`; the queried quantities are then merged onto the CG_4 and control catalogue rows by their stored `objid`.

The full query is provided in `src/data_loader.py` in the repository release (Zenodo DOI; Sect. Data availability).

Appendix B: sSFR classification details

We identify BPT-classifiable galaxies by requiring measured values of both emission-line ratios

$$\log_{10}([\mathrm{N\,II}]\lambda6584/\mathrm{H}\alpha) \quad \text{and} \quad \log_{10}([\mathrm{O\,III}]\lambda5007/\mathrm{H}\beta).$$

Among these galaxies, we flag AGN-like systems if they satisfy either

$$\log_{10}\frac{[\mathrm{N\,II}]}{\mathrm{H}\alpha} > 0,$$

or if they lie above the empirical demarcation of Kauffmann et al. (2003a):

$$\log_{10}\frac{[\mathrm{O\,III}]}{\mathrm{H}\beta} > \frac{0.61}{\log_{10}([\mathrm{N\,II}]/\mathrm{H}\alpha) - 0.05} + 1.3,$$

in which case they are excluded from the GMM calibration sample. This calibration flag is deliberately more inclusive than the three-class BPT scheme of Sect. 3.7, in which composite galaxies lie between the Kauffmann et al. (2003a) and Kewley et al. (2001) demarcations and only galaxies above the Kewley et al. (2001) line are labelled AGN-like, because its sole purpose is to keep possible AGN contamination out of the sSFR calibration. Galaxies with missing diagnostic line ratios are BPT-unclassified, not proven non-AGN. They are retained for the broad sSFR status assignment when valid SDSS stellar mass and sSFR estimates are available, while the emission-line/AGN-like analysis treats BPT-unclassified objects separately. Galaxies with `specsfr_tot_p50` equal to −9999 in the SDSS carry the MPA-JHU missing-value sentinel and have no valid sSFR estimate; they are treated as missing data throughout: excluded from the GMM calibration, never assigned an sSFR class, omitted from every sSFR figure, and reported as counts only (Sect. 2.2). Galaxies with a measured sSFR are divided into quenched and star-forming classes with a two-component Gaussian Mixture Model (GMM) in the $\log M_\star$-$\log \mathrm{sSFR}$ plane (McLachlan & Peel 2000). The calibration excludes galaxies flagged as AGN-like where the required BPT line ratios are available, while BPT-unclassified galaxies with valid SDSS stellar mass and sSFR estimates can still receive a broad sSFR class. The quenched/star-forming boundary is the locus where the two component posterior probabilities are equal; the resulting classification for the SDSS selection is shown in Fig. B.1. A fixed-threshold robustness check confirms that the qualitative hierarchy is not tied to the exact GMM split (Sect. 3.7).

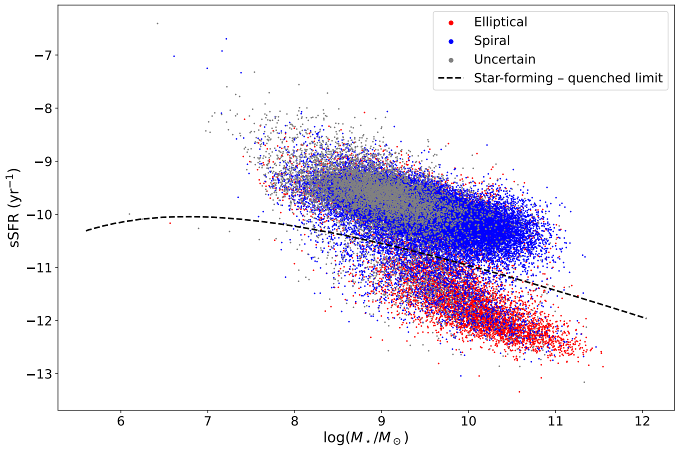

Fig. B.1. sSFR versus stellar mass, the star-forming/quenched limit, and Galaxy Zoo morphologies for the SDSS selection, excluding galaxies without sSFR estimates.

Appendix C: Compact groups as projected cores of ordinary groups

The compact-group exclusion of Sect. 2.1 is not a technicality: for 60 of the 765 parent Lim groups, the would-be $\mathrm{Control}_{4C}$ quartet (BGG plus three closest projected companions) contains at least one CG_4 galaxy — 192 CG_4 galaxies in total. Table C.1 breaks this overlap down by the Zheng & Shen (2021) embeddedness class: essentially every Embedded and Predominant compact group *is* the projected core of an ordinary group. Had these groups not been excluded, the same galaxies would have appeared on both sides of the comparisons, biasing every contrast toward null.

Appendix D: Star-forming main-sequence details

We model the star-forming main sequence as a function of stellar mass using polynomial orders 1–4; the polynomial-order comparison is provided in the online supplement. The diagnostic shown there is the residual RMS scatter, not a formal reduced χ^2, because no per-galaxy uncertainty model is used. We adopt order 2 because it is the lowest-order model with the smallest five-fold cross-validated RMS scatter (0.30523 dex, compared with 0.30560 dex for order 1). We then use order 2 to compute the residual of each star-forming galaxy with respect to the main sequence. A positive residual means higher sSFR at fixed stellar mass relative to the fitted star-forming main sequence.

Table C.1. CG_4 galaxies inside would-be $\mathrm{Control}_{4C}$ quartets of the parent catalogue, by compact-group class.

CG_4 class	CG_4 groups	Lim groups	Galaxies
Embedded	37 / 37	37	116
Isolated	6 / 6	6	24
Predominant	13 / 19	13	40
Split	4 / 16	4	12

Notes. "CG_4 groups" counts compact groups with at least one member inside a parent-group quartet, over the total number of compact groups of that class (split groups included, as in the exclusion).

In the comparisons below, Δ is the CG_4-minus-control median residual; positive values mean that CG_4 star-forming galaxies lie above the fitted main sequence relative to the control. Each quoted interval is the central 68% (16th–84th percentile) bootstrap interval on the median difference, and each p is the two-sided bootstrap sign-crossing probability for the same difference (10 000 resamples). The 68% interval and the two-sided p answer slightly different questions and can therefore appear to disagree near the significance boundary: a 68% interval that excludes zero is compatible with p slightly above 0.05.

The median main-sequence residual of CG_4 galaxies is:

- marginally different from that of Control4B, with a median offset of $\Delta = 0.056$ ($0.028 \leq \Delta \leq 0.098$, 16th–84th percentile bootstrap interval), corresponding to $p = 0.056$.
- significantly different from that of Control4C, with a median offset of $\Delta = 0.087$ ($0.067 \leq \Delta \leq 0.128$, 16th–84th percentile bootstrap interval), corresponding to $p = 0.009$.
- not significantly different from that of RG4, with a median offset of $\Delta = 0.046$ ($-0.037 \leq \Delta \leq 0.098$, 16th–84th percentile bootstrap interval), corresponding to $p = 0.532$.

Because the raw residual comparison and the matched comparison condition on different covariate structures, and because the apparent offset changes in strength after matching, we treat SFMS residuals as a secondary diagnostic rather than as evidence for a robust compact-group star-formation offset (Sect. 3.1).

Within CG_4, the star-forming main-sequence offsets of the Embedded, Isolated and Predominant classes remain statistically consistent with one another.

Appendix E: Environmental, group-scale, and colour diagnostics

This appendix collects the group-scale and colour diagnostics summarized in Sect. 3.7.

E.1. Distance to the BGG

We compare the sSFR of satellite galaxies with their distance to the BGG, expressed both in projected kpc and in units of $\langle R_{ij} \rangle$.

Significant monotonic trends are detected in the following cases.

In CG_4, sSFR correlates with projected distance to the BGG with Spearman coefficient $\rho_S = 0.24$ for 174 satellites ($p = 1.5 \times 10^{-3}$) and Pearson coefficient $r = 0.21$ ($p = 5.5 \times 10^{-3}$).

In $Control_{4B}$, sSFR correlates with distance to the BGG normalized by $\langle R_{ij} \rangle$ with Spearman coefficient $\rho_S = 0.08$ for 1972 satellites ($p = 3.7 \times 10^{-4}$) and Pearson coefficient $r = 0.08$ ($p = 2.4 \times 10^{-4}$).

In $Control_{4C}$, sSFR correlates with projected distance to the BGG with Spearman coefficient $\rho_S = 0.27$ for 1992 satellites ($p < 10^{-4}$) and Pearson coefficient $r = 0.23$ ($p < 10^{-4}$).

In $Control_{4C}$, sSFR correlates with distance to the BGG normalized by $\langle R_{ij} \rangle$ with Spearman coefficient $\rho_S = 0.10$ for 1992 satellites ($p < 10^{-4}$) and Pearson coefficient $r = 0.10$ ($p < 10^{-4}$).

E.2. Dominance

We split groups into dominated systems, where the BGG contributes more than 60% of the group luminosity, and non-dominated systems. This split does not produce a strong compact-group signal by itself: CG_4 shows no significant dominated/non-dominated difference after the Benjamini–Hochberg false-discovery-rate (FDR) correction. Here $p_{\rm adj}$ denotes the FDR-adjusted p-value, and Cliff's δ is a rank-based effect size ranging from -1 to 1.

Across the control and reference samples, the clearest binary separation is instead in group luminosity. Dominated systems have lower $L_{\rm group}$ than their non-dominated counterparts: $Control_{4B}$ gives 9.1×10^{10} versus 1.28×10^{11} $L_\odot$ ($p_{\rm adj} < 10^{-4}$, $\delta = -0.55$); $Control_{4C}$ gives 8.73×10^{10} versus 9.98×10^{10} $L_\odot$ ($p_{\rm adj} < 10^{-4}$, $\delta = -0.32$); and RG_4 gives 7.09×10^{10} versus 9.82×10^{10} $L_\odot$ ($p_{\rm adj} = 1 \times 10^{-4}$, $\delta = -0.59$). The negative δ values mean that dominated groups tend to occupy the lower-luminosity side of each comparison.

The spiral-fraction correlations show a broader environmental pattern rather than a clean dominated/non-dominated dichotomy. The most coherent set appears in $\mathrm{Control}_{4\mathrm{C}}$, where $S_{\rm frac}$ increases with $\left\langle R_{ij} \right\rangle$ in both dominated and non-dominated groups ($\rho_S = 0.40$, $p < 10^{-4}$ and $\rho_S = 0.38$, $p < 10^{-4}$), while weaker negative trends are found with σ_v and $L_{\rm group}$. $\mathrm{Control}_{4\mathrm{B}}$ contributes only weaker non-dominated-group correlations. In CG_4, the significant trends occur in small subsets, with $S_{\rm frac}$ decreasing with $L_{\rm group}$ among dominated groups ($\rho_S = -0.50$, 36 groups) and with $\Delta V_{\rm BGG}/\sigma_v$ among non-dominated groups ($\rho_S = -0.53$, 26 groups); we therefore treat these as supporting diagnostics rather than the main dominance result. The full distributional diagnostics are provided in the online supplement.

E.3. Morphology and domination

When we further split the samples by domination and by the morphology of the BGG, significant effects only appear in the following cases.

- $Control_{4B}$ dominated groups, where the median satellite spiral fraction is 1.00 for spiral BGGs and 0.67 for elliptical BGGs. The association between BGG morphology and satellite spiral fraction has a group-level odds ratio of 1.842 ($p = 0.01$) in this sample; the same within-sample contrast under a satellite-level clustered uncertainty treatment gives an odds ratio of 1.842 ($p = 0.02$).
- $Control_{4C}$ dominated groups, where the median satellite spiral fraction is 0.67 for spiral BGGs and 0.50 for elliptical BGGs. The association between BGG morphology and satellite spiral fraction has a group-level odds ratio of 1.918 ($p < 10^{-4}$) in this sample; the same within-sample contrast under a satellite-level clustered uncertainty treatment gives an odds ratio of 1.918 ($p < 10^{-4}$).

In each case the group-level and satellite-level clustered treatments share the same point estimate by construction and differ only in the uncertainty model, which is why the odds ratios coincide.

Table E.1. Elliptical and spiral counts by Zheng–Shen class for the CG_4 sample. Uncertain morphologies are excluded. Split systems have zero counts because they were removed by the existing non-split CG_4 sample definition.

Zheng–Shen class	All CG_4 E/Sp	CG_4 BGG E/Sp	CG_4 sat. E/Sp
Split	0/0	0/0	0/0
Isolated	7/11	3/1	4/10
Predominant	41/22	11/5	30/17
Embedded	76/53	22/11	54/42

Table E.2. Tests of the E-vs-Sp morphology mix against compact-group class, magnitude gap, and BGG luminosity fraction. The continuous columns report the available adjusted logistic model for $P(\mathrm{E})$; all-galaxy and satellite fits use group-clustered standard errors. Odds ratios are reported per magnitude for Δm_{12} and per 0.1 increase in luminosity fraction for $f_{L,\mathrm{BGG}}$.

Subset	Sample	Test	Result
all galaxies	CG_4	Class	$p = 0.137$
		Δm_{12}	OR 1.42, $p = 0.167$
		$f_{L,\mathrm{BGG}}$	OR 1.29, $p = 0.080$
all galaxies	RG_4	Class	not defined
		Δm_{12}	OR 1.33, $p = 0.220$
		$f_{L,\mathrm{BGG}}$	OR 1.09, $p = 0.540$
BGGs only	CG_4	Class	$p = 1.000$
		Δm_{12}	OR 1.77, $p = 0.426$
		$f_{L,\mathrm{BGG}}$	OR 1.68, $p = 0.174$
BGGs only	RG_4	Class	not defined
		Δm_{12}	OR 1.58, $p = 0.312$
		$f_{L,\mathrm{BGG}}$	OR 1.21, $p = 0.520$
satellites only	CG_4	Class	$p = 0.066$
		Δm_{12}	OR 1.28, $p = 0.346$
		$f_{L,\mathrm{BGG}}$	OR 1.18, $p = 0.321$
satellites only	RG_4	Class	not defined
		Δm_{12}	OR 1.41, $p = 0.342$
		$f_{L,\mathrm{BGG}}$	OR 1.12, $p = 0.584$

E.4. Morphology versus Zheng–Shen class, magnitude gap, and BGG dominance

We next test whether the E/Sp morphology mix is associated with three group-level descriptors: the Zheng–Shen compact-group class, the first–second magnitude gap $\Delta m_{12} = M_{r,2} - M_{r,1}$, and the continuous BGG luminosity fraction $f_{L,\mathrm{BGG}} = L_{\mathrm{BGG}}/L_{\mathrm{group}}$. These tests use only the existing Galaxy Zoo elliptical and spiral classes; uncertain morphologies are excluded from the binary fits. We run the tests separately for all galaxies, BGGs only, and satellites only, so that any trend driven by the BGG itself is not folded into the satellite result. For satellites, $f_{L,\mathrm{BGG}}$ mechanically includes the satellite luminosity in the group-luminosity denominator, so any satellite-level trend with $f_{L,\mathrm{BGG}}$ is partly definitional; we therefore report the satellite models with a stellar-mass control.

The magnitude-gap and BGG-luminosity-fraction quantities are available for all 118 CG_4+RG_4 groups used in this test. The Zheng–Shen class is available only for compact groups. Split systems are absent from the current CG_4 sample because they are removed before the paper analysis, and no corresponding Zheng–Shen class is defined for the RG_4 catalogue used here. We therefore report the RG_4 class test as unavailable, while retaining RG_4 in the Δm_{12} and $f_{L,\mathrm{BGG}}$ comparisons.

Overall, these tests provide little evidence that the E/Sp mix is driven by Zheng–Shen class, magnitude gap, or BGG luminosity fraction. The few marginal associations are therefore treated as descriptive diagnostics rather than as robust secondary trends (Tables E.1 and E.2).

For the Zheng–Shen compact-group class, no CG_4 association reaches the conventional 5% significance threshold. The closest case is the satellite-only test, which is treated as marginal and descriptive. Because the Zheng–Shen class is not defined for the RG_4 catalogue, we do not attempt an RG_4 class comparison.

For Δm_{12}, no statistically significant morphology trend is found in CG_4 or RG_4, whether the fit is run for all galaxies, BGGs only, or satellites only. The same conclusion holds for $f_{L,\mathrm{BGG}}$: no fitted trend reaches $p < 0.05$. The CG_4–RG_4 interaction models detect no significant difference in the morphology–dominance relation for either Δm_{12} or $f_{L,\mathrm{BGG}}$, across all galaxies, BGGs only, and satellites only. A class interaction is not defined because the Zheng–Shen compact-group class has no RG_4 counterpart in the present catalogue. Across the morphology–class, morphology–gap, morphology–luminosity-fraction, and interaction tests, no result remains significant after Benjamini–Hochberg FDR control.

E.5. Crossing time

The crossing-time diagnostic mainly separates systems by dynamical scale rather than by luminosity alone. Shorter t_{cross} systems have larger virial mass-to-light ratios and virial masses: the correlations with $\log(\mathcal{M}_{\mathrm{VT}}/L_r)$ and $\log(\mathcal{M}_{\mathrm{VT}})$ are $\rho_{\mathrm{S}} = -0.39$ ($p < 10^{-4}$) and $\rho_{\mathrm{S}} = -0.33$ ($p < 10^{-4}$), respectively, over 816 groups. By contrast, the relation with total group luminosity is formally detected but very weak ($\rho_{\mathrm{S}} = 0.08$). We therefore read t_{cross} as a dynamical compactness indicator tied primarily to virial mass and mass-to-light ratio, with little independent leverage from L_{group}.

E.6. Optical colours

The retained colour-analysis matches include 110 of 248 galaxies in CG_4, 931 of 2792 galaxies in $\mathrm{Control}_{4B}$, 1368 of 2816 galaxies in $\mathrm{Control}_{4C}$, 102 of 224 galaxies in RG_4. This stricter subset differs from the availability audit in Fig. F.2: the figure's colour row only requires the four derived SDSS colour columns to be present in the final catalogue, giving 217 of 248 CG_4 galaxies, whereas the colour analysis also requires successful matched-photometry retention and the covariates used below. The fractions retained in this stricter colour-analysis subset are 44.4% in CG4, 33.3% in Control4B, 48.6% in Control4C, 45.5% in RG4.

The raw catalogue colour–mass fits indicate that CG_4 does not simply follow the same colour–mass relation as every comparison sample: common slopes are rejected in all four colours, with the significant directional slope differences concentrated in the bluer colours and dependent on the adopted comparison catalogue. In the satellite-only comparison, the fitted offsets and slopes likewise depend on the control definition. These tests motivate the robustness analysis rather than establishing a stand-alone compact-group colour effect.

We therefore refit the satellite colours with cluster-robust standard errors, using catalogue-qualified group identifiers. In the pooled comparison adjusted only for stellar mass and redshift, none of the four CG_4 coefficients is significantly negative after Holm correction. Adding morphology and sSFR class does not reveal a significant negative coefficient. However, these adjusted coefficients are control-sample dependent, and they do not persist within the sSFR or morphology splits. The data therefore do not localize the colour effect to the star-forming spiral population expected for a simple interaction-triggered star-formation scenario.

Residual colour trends with projected BGG-centric distance are consistent with stronger processing near the group centre and a less processed outer population, but these trends use the same biased colour subset and are treated as exploratory. The group-level BGG–satellite colour concordance is similarly secondary: redder BGGs tend to have redder mean satellite populations in the combined group sample, but the CG_4 slope is not a robustly distinct population-level signature. Detailed colour–mass, satellite, and radial diagnostics are provided in the online supplement. Figure E.1 summarizes the model and control-sample dependence that motivates our interpretation: the present data do not establish a robust global blue or red offset for CG_4 satellites.

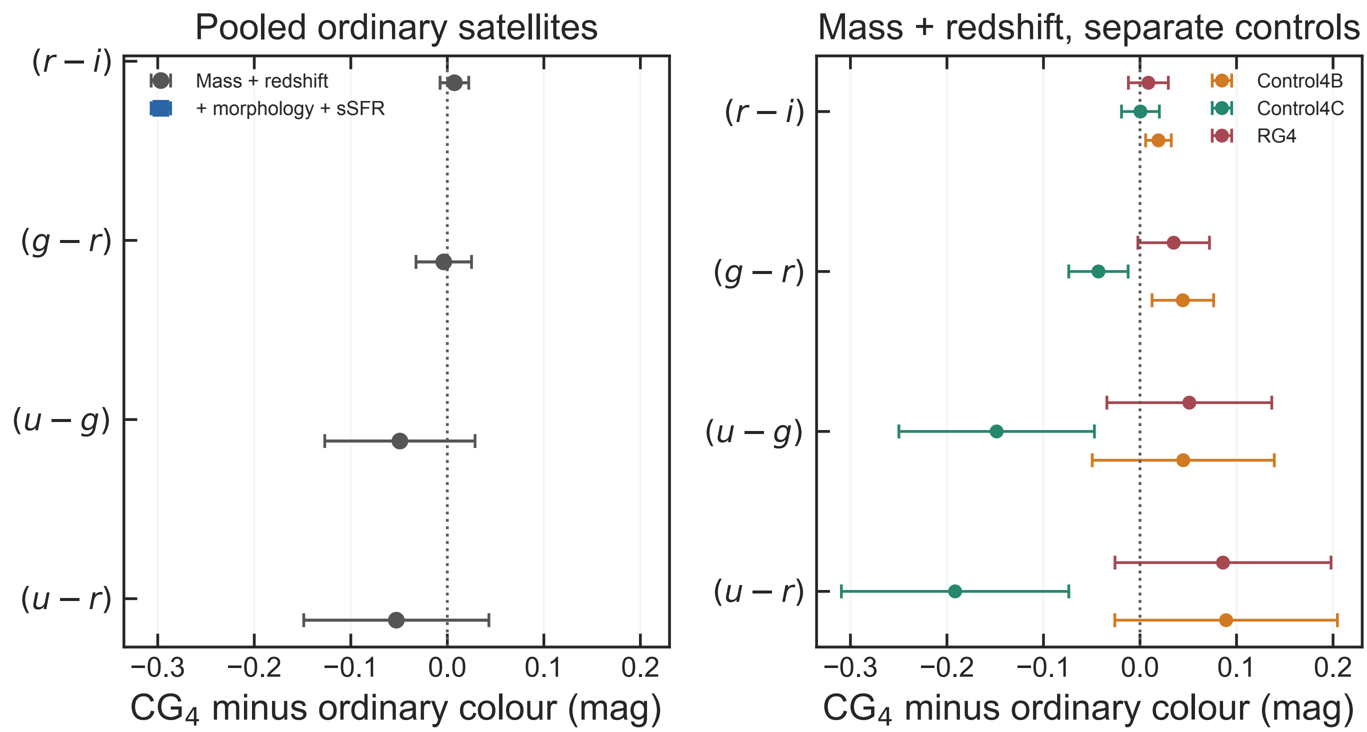

Fig. E.1. Robustness of the satellite colour coefficients. Left: pooled ordinary-group comparison before and after controlling for morphology and sSFR class. Right: mass- and redshift-adjusted comparisons against each control catalogue separately. Points show the CG_4 minus ordinary coefficient and bars show 95% confidence intervals; negative values indicate bluer CG_4 satellites.

Appendix F: Pooled models, matching details, and selection diagnostics

F.1. Group-level multiplicity sensitivity

The three per-control group-level satellite-composition matches are treated in the main text as distinct control-definition estimands, and their p-values are not adjusted across $\mathrm{Control}_{4B}$, $\mathrm{Control}_{4C}$, and RG_4. Table F.1 reports the Holm sensitivity requested by that alternative convention. The pooled group-level match is excluded because it is a secondary summary of overlapping control catalogues.

F.2. Pooled adjusted models

As a compact summary across the heterogeneous control definitions we also fit binomial models to the pooled catalogue. Because the control samples overlap heavily (all RG_4 galaxies are also $\mathrm{Control}_{4B}$ members, and $\mathrm{Control}_{4B}$ and $\mathrm{Control}_{4C}$ share their cores), the pooled control side is first deduplicated by SDSS objid — one row per physical galaxy, unweighted, with every source label recorded — and standard errors are clustered by physical Lim group. The common adjustment set includes stellar mass, redshift, satellite/BGG rank, group luminosity and velocity dispersion when sufficiently complete. This pooled family is read as a secondary summary; the per-control contrasts of Sect. 3.2 are primary.

Table F.1. Holm sensitivity for the three per-control group-level satellite-composition matches. The adjusted values are computed from the full-precision bootstrap and paired sign-flip permutation p-values.

Control	p_{perm}	Holm p_{perm}	p_{boot}	Holm p_{boot}
$\mathrm{Control}_{4B}$	0.016	0.047	0.012	0.036
$\mathrm{Control}_{4C}$	0.027	0.053	0.021	0.038
RG_4	0.026	0.053	0.019	0.038

For quenched status, the CG_4 odds ratio is 1.08, with 95% confidence interval $[0.80, 1.44]$ and Holm-corrected $p = 1.000$.

For quenched satellite status, the CG_4 odds ratio is 1.17, with 95% confidence interval $[0.86, 1.58]$ and Holm-corrected $p = 1.000$.

For star-forming satellite status, the CG_4 odds ratio is 0.86, with 95% confidence interval $[0.63, 1.16]$ and Holm-corrected $p = 1.000$.

For elliptical morphology, the CG_4 odds ratio is 1.50, with 95% confidence interval $[1.13, 1.99]$ and Holm-corrected $p = 0.032$.

For elliptical satellite morphology, the CG_4 odds ratio is 1.71, with 95% confidence interval $[1.24, 2.34]$ and Holm-corrected $p = 0.007$.

For elliptical BGG morphology, the CG_4 odds ratio is 0.77, with 95% confidence interval $[0.39, 1.53]$ and Holm-corrected $p = 1.000$.

For spiral morphology, the CG_4 odds ratio is 0.67, with 95% confidence interval $[0.50, 0.89]$ and Holm-corrected $p = 0.032$.

For spiral satellite morphology, the CG_4 odds ratio is 0.59, with 95% confidence interval $[0.43, 0.80]$ and Holm-corrected $p = 0.007$.

Compact-group membership therefore remains independently associated with morphology after the available adjustments: CG_4 galaxies have a higher probability of being elliptical and a lower probability of being spiral. In contrast,

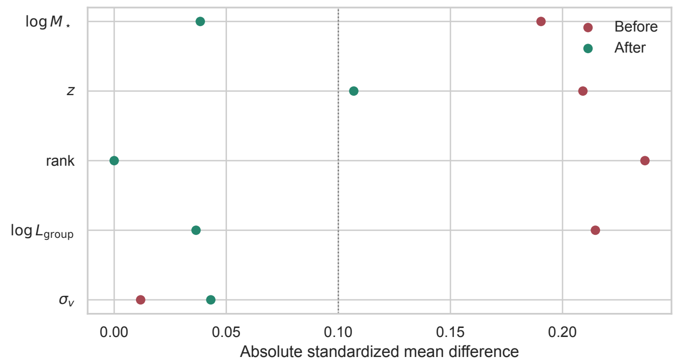

Fig. F.1. Absolute standardized mean differences before and after matching.

the adjusted quenched and star-forming satellite coefficients are weaker and do not provide evidence for an instantaneous star-formation difference that remains significant after correction.

F.3. Galaxy-level matching details

The galaxy-level control pool of Sect. 3.3 is deduplicated by SDSS objid before matching: 5832 control rows collapse to 3983 unique galaxies. The 14 unmatched CG_4 galaxies were excluded because they lay outside the common propensity-score support or exceeded the adopted caliper. By kept label, the matched controls comprise 141 Control4B, 77 Control4C, 16 RG4; by physical provenance, 16 of the 234 matched controls are physically RG_4 galaxies (which also appear in $\mathrm{Control}_{4B}$ through the sample overlap), and the full per-galaxy provenance is recorded in the released `matched_control_provenance.csv`. The covariate balance before and after matching is shown in Fig. F.1; the per-outcome matched-effect figure is provided in the online supplement.

For completeness, the complement outcomes of the matched family are as follows.

The CG_4–minus–control difference in star-forming fraction is 0.022, with 95% bootstrap interval $[-0.055, 0.100]$ and Holm-corrected $p = 1.000$.

The CG_4–minus–control difference in spiral fraction is -0.087, with 95% bootstrap interval $[-0.179, 0.005]$ and Holm-corrected $p = 0.361$.

F.4. Selection diagnostics

The fractions with complete measurements for the major derived quantities are shown in Fig. F.2. The colour-matched versus unmatched comparisons quantify shifts in stellar mass, redshift, rank, star-formation class and morphology.

The median nearest-neighbour angular separation is 100.9 arcsec in CG_4 and 210.5 arcsec in the controls; the fractions below the 55-arcsec SDSS fibre-collision scale are 19.4% and 9.2%, respectively.

Under the availability definitions shown in Fig. F.2, no major availability difference meets both criteria of corrected significance and a difference exceeding 10 percentage points.

The group-scale availability row is based on the group covariates available for this catalogue: radius scale, velocity dispersion, group luminosity, and BGG luminosity dominance. A harmonized group-mass estimate is not available in these tables, so models that request it omit that covariate.

The colour-selection mass audit is provided in the online supplement.

Fig. F.2. Availability fractions for the major derived quantities in each sample. The colour row records completeness of the four derived SDSS colour columns in the final catalogue, rather than the stricter colour-analysis matched subset used in Appendix E.6.

Appendix G: Extended size diagnostics

The descriptive mass–size relations and the per-sample sizes predicted at $\log(M_\star/M_\odot) = 10.3$ are provided in the online supplement.

The crowding checks mirror Sect. 3.5: excluding galaxies with a projected neighbour within 55 arcsec leaves a satellite coefficient of 0.009 dex ($p = 0.625$), and adding the close-neighbour flag as a covariate gives 0.010 dex ($p = 0.542$).

The per-galaxy difference between the two measures, $d = \log R_{\rm chl,r} - \log R_{50,r}$, averages 0.045 dex, and is larger for elliptical galaxies (0.063 dex) than for spiral galaxies (0.037 dex), as expected if the Petrosian aperture preferentially truncates concentrated profiles. Its rank correlation with the nearest-neighbour separation is $\rho_S = 0.064$ ($p = 3.1 \times 10^{-4}$), so residual blending does not drive a strong measure-dependent size systematic.

Adding the pairwise projected tidal index of Sect. 3.6 to the all-galaxy size model changes the CG_4 coefficient from 0.006 dex to 0.022 dex on the pairwise complete-case sample; with a near-null baseline coefficient, the attenuation diagnostic used for morphology is not informative here.

The radial diagnostic is exploratory: mass- and redshift-adjusted satellite size residuals correlate with projected BGG-centric distance with $\rho_S = 0.18$ ($p = 0.022$) in CG_4 and $\rho_S = 0.21$ ($p < 10^{-4}$) in the pooled controls. A more negative inner-region residual would be the tidal-truncation expectation; the observed trends are similar in CG_4 and the controls, so the data do not isolate a CG_4-specific inner-region size suppression.

Within morphology classes, the satellite-only fits give 0.038 dex for elliptical satellites (Holm-corrected $p = 0.151$) and 0.009 dex for spiral satellites (Holm-corrected $p = 0.724$), and the CG_4×morphology interaction is 0.035 dex ($p = 0.297$). These strata are secondary diagnostics; they test whether any size offset is independent information beyond the morphology signal rather than a restatement of it.

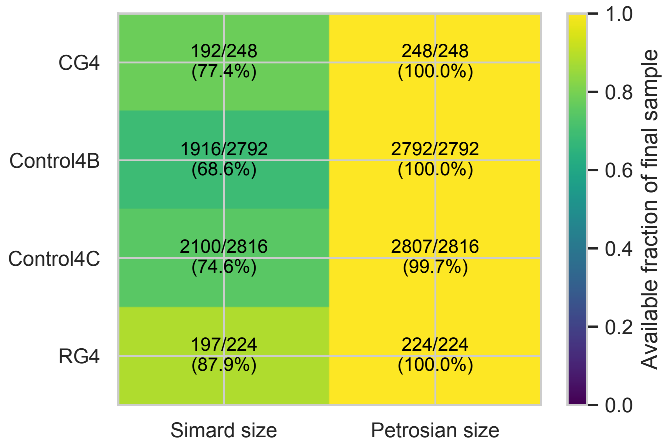

Fig. G.1. Availability of the seeing-corrected Simard half-light radius and the Petrosian half-light radius in each sample, after the bridge, quality, and blending cuts of Sect. 2.3.

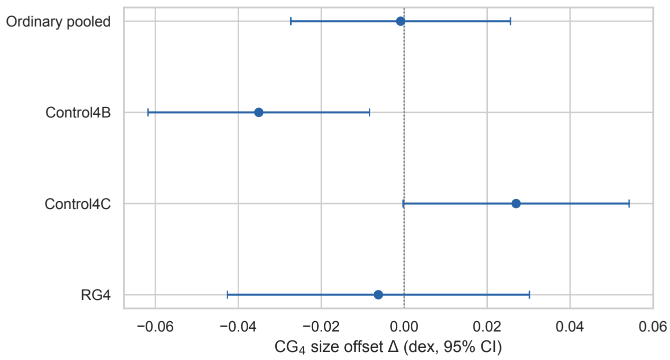

Fig. G.2. Mass- and redshift-adjusted CG_4 size coefficients against the pooled controls and against each control catalogue separately. Bars show 95% confidence intervals; negative values mean smaller CG_4 sizes at fixed mass.

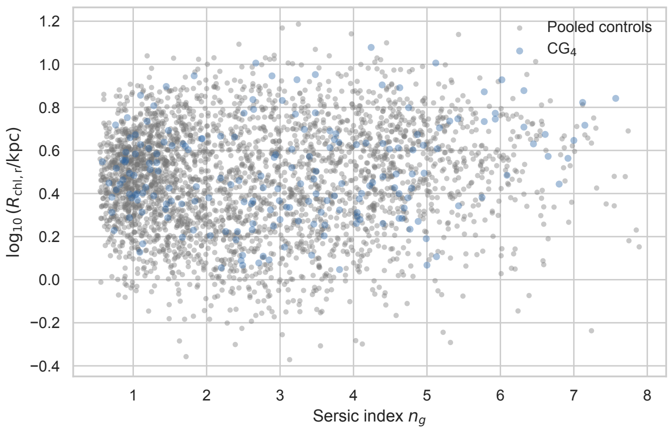

Fig. G.3. Half-light radius versus Sérsic index for CG_4 and the pooled controls, shown for galaxies with both valid Simard size and stellar mass in the mass–size analysis. This is the plane used by the S-PLUS transition-galaxy studies.